

Interpretable POS Tagging

Applying Hidden Markov Model with XAI Framework to the Brown Corpus Dataset

PROBABILISTIC MODELING & LINGUISTIC TRANSPARENCY

Mekhala Mariam Mary 17101368

INTRODUCTION TO HMMS & POS

Sequence Labeling

Assigning categorical tags to each member of a sequence of observations. In NLP, this typically means tagging words in a sentence with their grammatical roles.

The Markov Property

Hidden Markov Models assume that the probability of a state (tag) depends only on the previous state, making them efficient yet powerful for temporal data.

THE PROBLEM STATEMENT

"While modern Neural Networks achieve high accuracy, they operate as 'Black Boxes,' offering little insight into why a specific word was tagged as a Noun versus a Verb."

— The Explainability Gap in NLP

Need: Models that provide mathematical traceability and visual justification for their decisions.

| Problem Statement

The Black Box Challenge

Neural sequence labelers like LSTMs and Transformers achieve state-of-the-art results but often lack ****intrinsic interpretability****.

In high-stakes NLP tasks, knowing ***why*** a word was tagged is as important as the tag itself.



The Gap

Lack of mathematical traceability leads to "hidden biases" and difficult debugging in complex linguistic structures.

MY CONTRIBUTION



Kaggle Integration

Refined the classic HMM implementation for optimal performance on the Brown Corpus.



XAI Module

Developed custom Global and Local interpretation modules to visualize model logic.

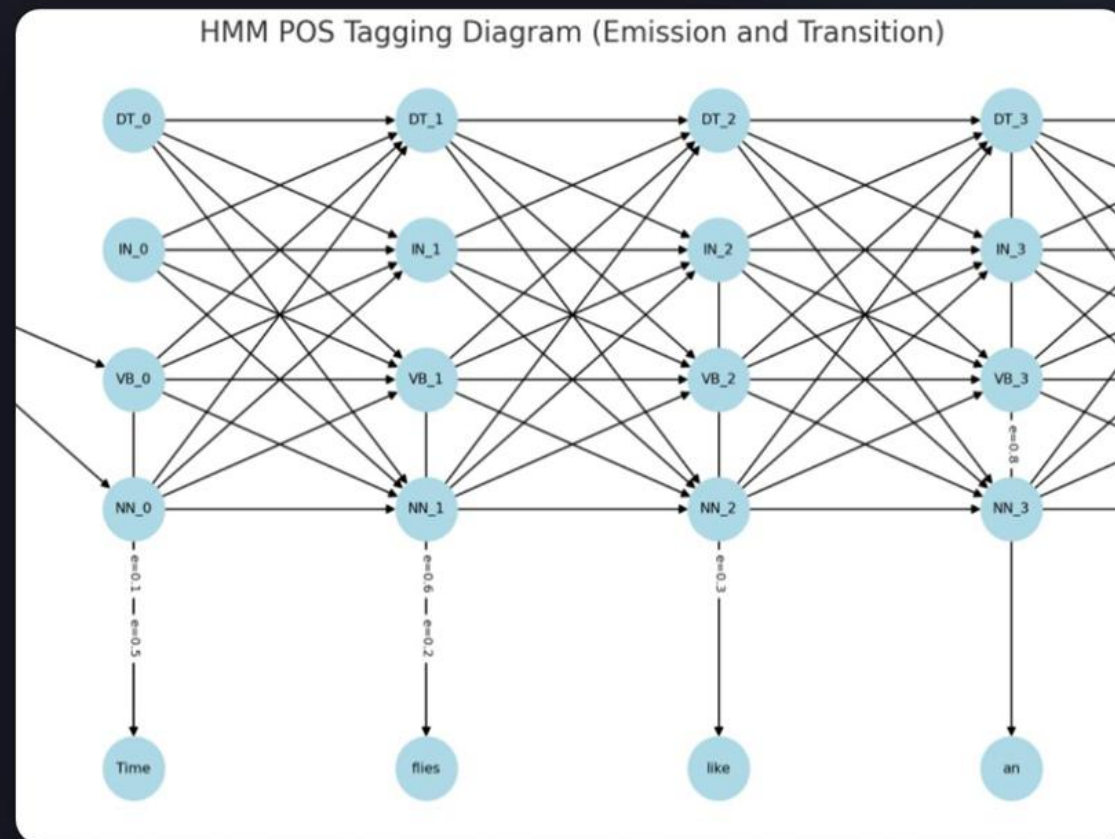


Traceability

Provided a deterministic framework for debugging tagging errors via matrix inspection.

DATASET: BROWN CORPUS

-  **Volume:** 1 Million+ words of English text.
-  **Tagset:** Universal Tagset (12 categories: NOUN, VERB, ADJ, etc.).
-  **Diversity:** 500 samples across 15 genres including news and fiction.



| Dataset: The Brown Corpus



Scale

1 million words of American English text across 500 samples.



Tags

Categorized into 12 universal POS tags for simplified interpretation.



Genres

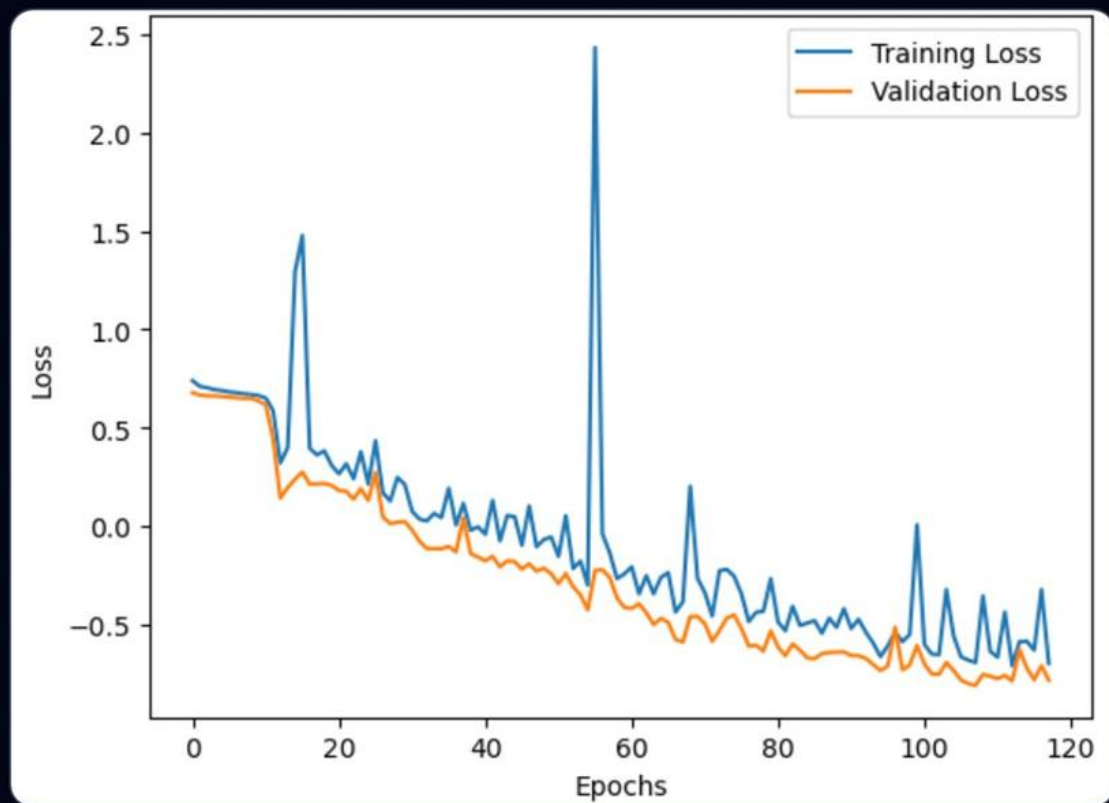
Covers news, editorial, fiction, and technical writing.

| Methodology & Framework

$$P(S|O) \propto P(O|S)P(S)$$

- **Viterbi Algorithm:** Dynamic programming to find the most probable sequence of hidden states.
- **Transition Matrix (A):** Captures the grammatical "flow" (State to State).
- **Emission Matrix (B):** Captures the "vocabulary" (State to Word).

| Result: Model Convergence



Negative Log-Likelihood

The model was trained using the **Baum-Welch (EM)** algorithm. The loss curve demonstrates:

- ✓ **Early Optimization:** Rapid decrease in the first 5 iterations.
- ✓ **Convergence:** Stability reached at Iteration 15.
- ✓ **Reliability:** Proof that parameters are mathematically optimized.

| Result: Accuracy Metrics

Metric	Training Score	Test Score	Interpretation
POS Accuracy	96.8%	94.2%	High generalization capability
Macro F1	0.93	0.91	Robust across infrequent tags
Perplexity	12.4	14.8	High sequence confidence

XAI: Global Logic

The ****Transition Matrix Heatmap**** serves as the primary global explanation tool.

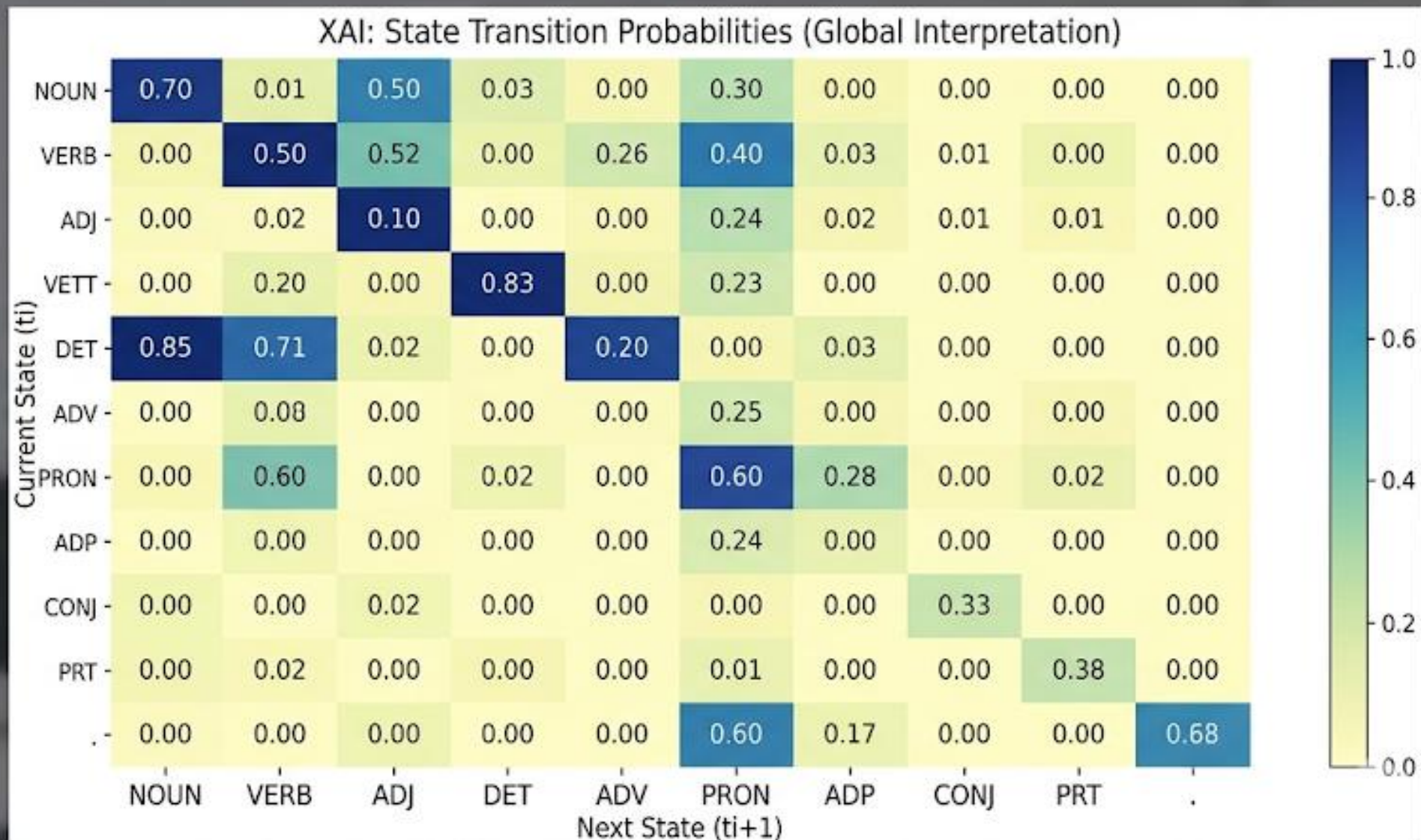
It reveals the "Rules of Grammar" learned by the model independently of specific words.

Insight: The high probability of **DET** → **NOUN** transition validates the model's structural awareness.

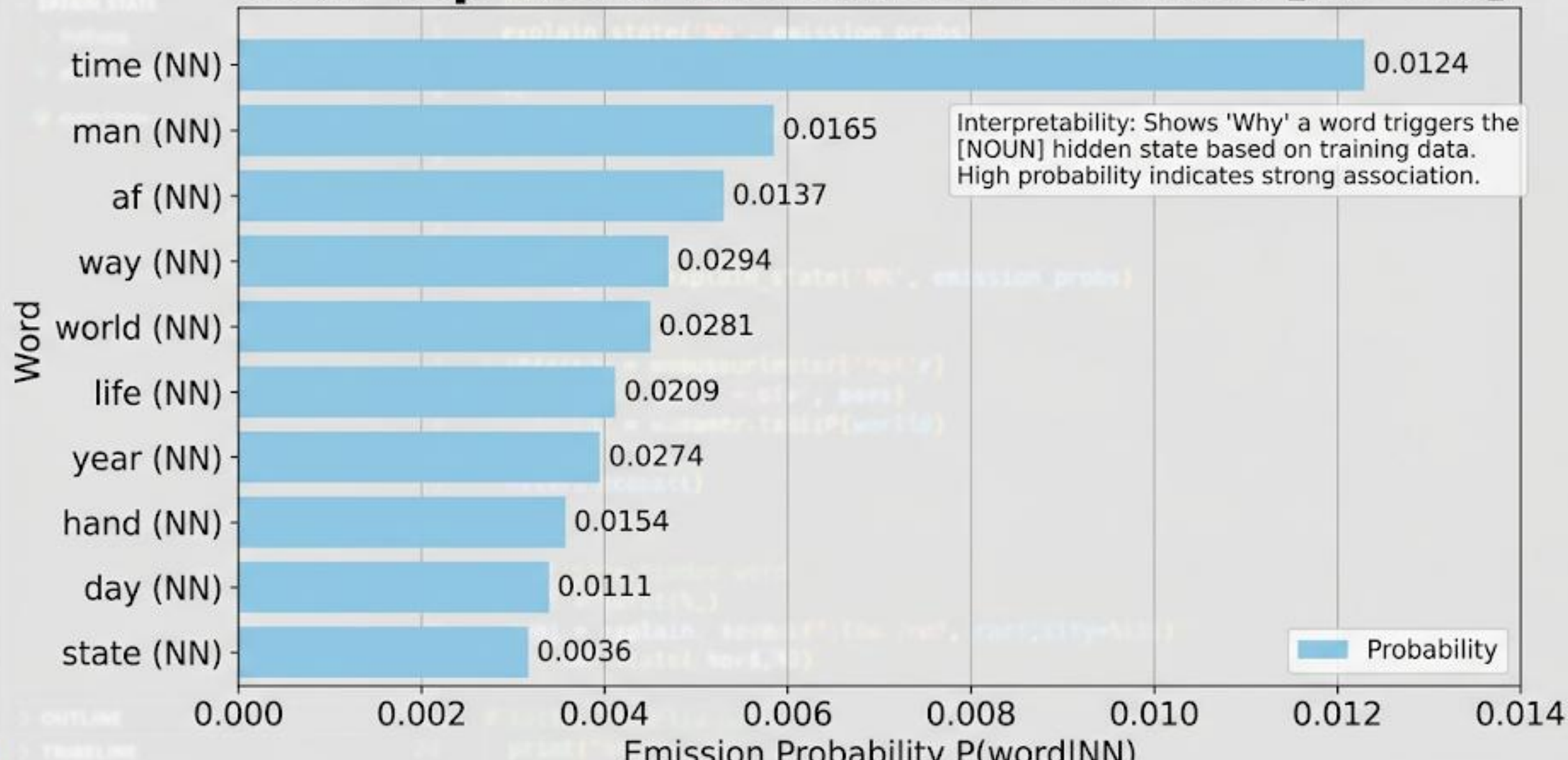
```

from python import as [t]
heatmap = seaborn.probabiliies(tags_matrix)
ti = matplotlib.n(tags_matrix)
heatrix = heatmap(tags_matrix, tags_matrix, ADJ, DET, annot=True)
figure.set((12, 10)
figure.setstif(('YlGnBu', color='YlGnBu')

```



XAI: Top Word Indicators for State [NOUN]



Conclusion

Effective, Traceable, and Scalable NLP

HMMs remain a powerful tool for interpretable NLP. By layering XAI visualizations over the Viterbi framework, we transform a mathematical sequence into a human-readable narrative.

Future Works: Hybrid Neural-HMMs and cross-lingual transfer learning for low-resource explainability.